CE Radio Test Report

APPLICANT	:	Magne AI Global tech limited
PRODUCT NAME	:	MAG1
BRAND NAME	:	MaQ
MODEL NAME	:	MA1
STANDARD	:	ETSI EN 303 413 V1.2.1 (2021-04)
TEST DATE(S)	:	Mar. 14, 2026 ~ May 19, 2026

The measurements shown in this test report were made in accordance with the procedures given in ETSI EN 303 413 V1.2.1 (2021-04) , and shown to be compliant with the applicable technical standards.
The test results in this report apply exclusively to the tested model / sample. Without written approval of Sporton International Inc. (KunShan), the test report shall not be reproduced except in full.

Jason Jia

Approved by: Jason Jia

ACCREDITED
Cert #5145.02

Sporton International Inc. (Kunshan)
No. 1098, Pengxi North Road, Kunshan Economic Development Zone Jiangsu Province 215300 People's Republic of China

TABLE OF CONTENTS

REVISION HISTORY

REPORT NO.	VERSION	DESCRIPTION	ISSUED DATE
ER612311C	Rev. 01	Initial issue of report	May 25, 2026

SUMMARY OF TEST RESULT

SECTION	CLAUSE (EN 303 413)	TEST PARAMETER	Result
3.1	4.2.1	Receiver Blocking	PASS
3.2	4.2.2	Receiver spurious emissions	PASS

Conformity Assessment Condition:

1. The test results (PASS/FAIL) with all measurement uncertainty excluded are presented against the regulation limits or in accordance with the requirements stipulated by the applicant/manufacturer who shall bear all the risks of non-compliance that may potentially occur if measurement uncertainty is taken into account.
2. The measurement uncertainty please refer to each test result in the section “Measurement Uncertainty”

Disclaimer:

The product specifications of the EUT presented in the test report that may affect the test assessments are declared by the manufacturer who shall take full responsibility for the authenticity.

1. General Description

1.1 Applicant

Magne AI Global tech limited
FLAT 1019B,10/F,LIVEN HOUSE,NO.61-63 KING YIP STREET KWUN TONG HK

1.2 Manufacturer

Magne AI Global tech limited
FLAT 1019B,10/F,LIVEN HOUSE,NO.61-63 KING YIP STREET KWUN TONG HK

1.3 Feature of Equipment Under Test

Product Feature & Specification	
Product Name	MAG1
Brand Name	MaQ
Model Name	MA1
IMEI Code	Conducted: 016813000001410/016813000001428 Receiver spurious emissions: 016813000001816

GNSS Constellation	**GNSS Signals**		
	GNSS Frequency Band (MHz)		
	1559-1610	1164-1215	1215-1300
BDS	■ B1I □ B1C		
Galileo	■ E1	□ E5a □ E5b	□ E6
GLONASS	□ G1		□ G2
GPS	■ L1 C/A ■ L1 C	□ L5	□ L2C
SBAS	□ L1	□ L5	

Remark:
The above EUT's information was declared by manufacturer. Please refer to the specifications or user's manual for more detailed description.

1.4 Modification of EUT

No modifications are made to the EUT during all test items.

1.5 Testing Facility

Sporton International Inc. (Kunshan) is accredited to ISO/IEC 17025:2017 by American Association for Laboratory Accreditation with Certificate Number 5145.02.

Test Site	Sporton International Inc. (Kunshan)
Test Site Location	No. 1098, Pengxi North Road, Kunshan Economic Development Zone Jiangsu Province 215300 People's Republic of China TEL : +86-512-57900158
Test Site No.	**Sporton Site No. :** DFS01-KS; 05CH02-KS

1.6 Test Software

Item	Site	Manufacturer	Name	Version
1.	05CH02-KS	AUDIX	E3	210616
2.	DFS01-KS	Sporton	Test Tools	1.0

1.7 Applied Standards

According to the specifications of the manufacturer, the EUT must comply with the requirements of the following standards:

- **ETSI EN 303 413 V1.2.1 (2021-04)**

Note : All test items were verified and recorded according to the standards and without any deviation during the test.

1.8 Description of Support Test System

Item	Equipment	Trade Name	Model Name	FCC ID	Data Cable	Power Cord
1.	Earphone	Lenovo	P121	N/A	N/A	Unshielded,1.2m
2.	GNSS Station	ADIVIC	MP9000	N/A	N/A	Unshielded,1.8m

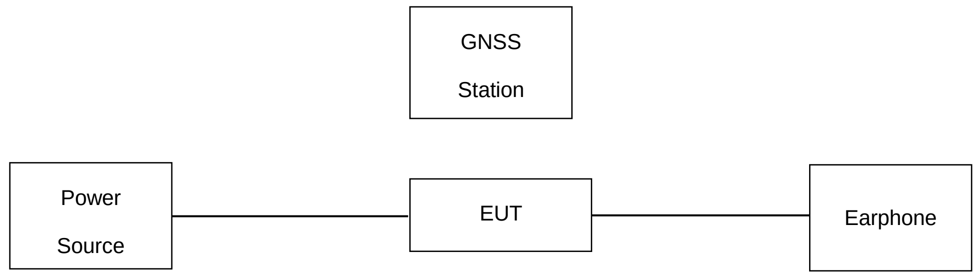
2. Test Configuration of Equipment Under Test

2.1 Test Mode

The following test mode was performed for Spurious Emissions:

Mode	Condition
1	GNSS RX(1559-1610MHz) + USB Cable (Charging from Adapter)+Earphone

2.2 Connection Diagram of Test System



This example is connection diagram of EUT test configurations.

. For detail, please refer to test mode configuration and setup photographs for each test item.

2.3 Test Configurations and Test Software

The EUT was set in receive GNSS signal, and used tool that C/N_0 value of GNSS can be reported by the EUT.

3. Conformance Specifications

3.1 Receiver Blocking

3.1.1 Requirement

The C/N_0 metric reported by the GUE for all GNSS constellations and GNSS signals given in table 4-1 and supported by the GUE shall not degrade by more than the value given in equation (4-1) when a blocking signal is applied. The blocking signal is defined in EN 303 413 table 4-4, with the frequencies and power levels defined in table 4-2 and/or in table 4-3 depending on the RNSS bands supported by the GUE.

Equation (4-1), Maximum degradation: $\triangle C/N0 \leq 1$ dB

EN 303 413 Table 4-1: GNSS constellations, GNSS signals and RNSS frequency bands

GNSS constellation	GNSS Signal Designations	RNSS Frequency Band (MHz)
BDS	B1I	1559 to 1610
	B1C	1559 to 1610
Galileo	E1	1559 to 1610
	E5a	1164 to 1215
	E5b	1164 to 1215
	E6	1215 to 1300
GLONASS	G1	1559 to 1610
	G2	1215 to 1300
GPS	L1 C/A	1559 to 1610
	L1C	1559 to 1610
	L2C	1215 to 1300
	L5	1164 to 1215
SBAS	L1	1559 to 1610
	L5	1164 to 1215

3.1.2 GNSS signal details

All GNSSs and GNSS signals declared as supported in the test report shall be simulated during the conformance testing. The relevant GNSSs and GNSS signals and the relative signal levels between signal types per GNSS are detailed in Table B-1.

Table B-1: Signal power levels for each GNSS signal within each GNSS constellation supported

GNSS constellation	GNSS signal	Signal power level(note)
BDS	B1I	-133 dBm
	B1C (IGSO)	-131 dBm
	B1C (MEO)	-129 dBm
Galileo	E1	-127 dBm
	E5a	-125 dBm
	E5b	-125 dBm
	E6	-125 dBm
GLONASS	G1	-131 dBm
	G2	-137 dBm
GPS	L1 C/A	-128.5 dBm
	L1C	-127 dBm
	L2C	-130 dBm
	L5	-124.9 dBm
SBAS	L1	-131 dBm
	L5	-127.5 dBm

Note: The signal power levels represent the total signal power of the satellite per channel, not for example pilot and data channels separately.

3.1.3 Blocking signal Parameter

EN 303 413 Table 4-4: Blocking signal

Parameter	Value	Comments
Frequency	As Table 4-2 and 4-3	
Power level	As Table 4-2 and 4-3	
Bandwidth	1 MHz	See below for details
Format	AWGN	

The 1 MHz filtered AWGN signal used for interferer simulation shall follow the requirements specified in this clause:

- The passband shall have a bandwidth of at least 1 MHz. The bandwidth is defined as the frequency range in which the attenuation of the filter is not more than 3 dB

- Within the passband, the ripple shall be less than 3 dB

- The stopband attenuation shall be at least 62 dB

- The transition band between the passband and stopband is 4 MHz wide

- Within the transition band, the filter shall not exceed the maximum passband gain.

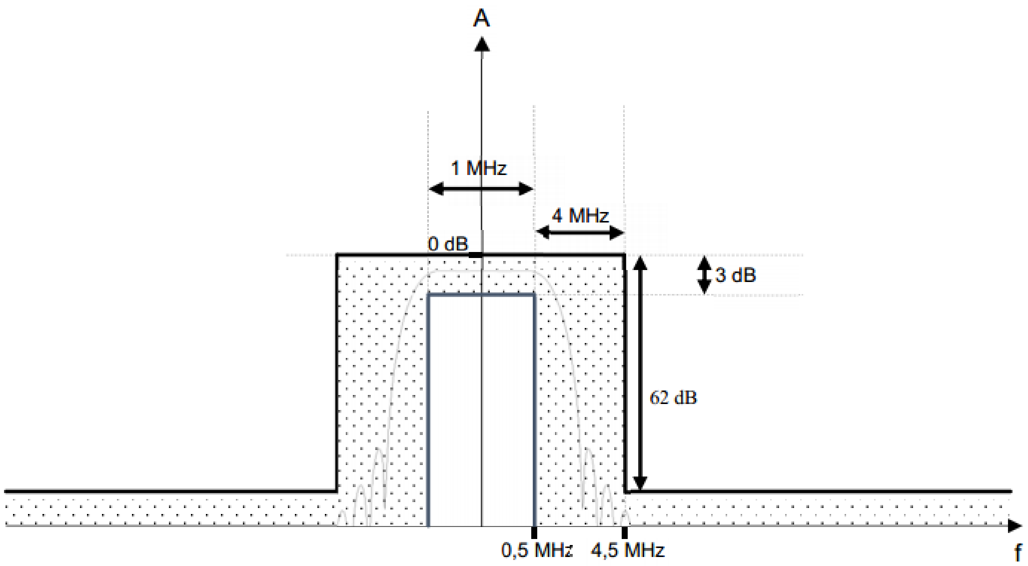


Figure B-1: Filter used to generate the blocking signal

EN 303 413 Table 4-2: Frequency bands, blocking signal test point frequencies and power levels for the 1559 MHz to 1610 MHz GNSS band

Frequency band (MHz)	Test point centre frequency (MHz)	Blocking signal power level (dBm)	Comments
1518- 1525	1524	-65	MSS (space-to-Earth) band
1525-1549	1548	-95	MSS (space-to-Earth) band
1549-1559	1554	-105	MSS (space-to-Earth) band
1559-1610	GNSS band under test		
1610-1626	1615	-105	MSS (Earth-to-space) band
1626- 1640	1627	-85	MSS (Earth-to-space) band

EN 303 413 Table 4-3: Frequency bands, blocking signal test point frequencies and power levels for the 1164 MHz to 1300 MHz GNSS band

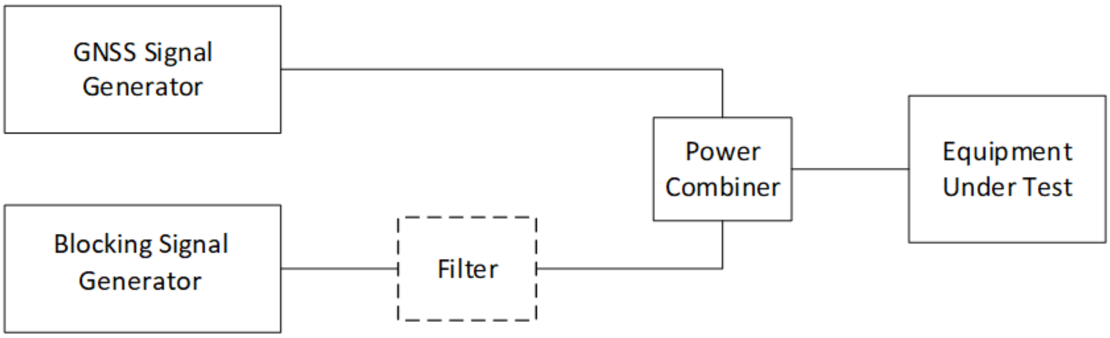
Frequency band (MHz)	Test point centre frequency (MHz)	Blocking signal power level (dBm)	Comments
960-1164	1154	-75	AM(R)S, ARNS band
1164-1215	GNSS band under test		
1215-1260	GNSS band under test		
1260-1300	GNSS band under test		
1300-1350	1310	-85	Radiolocation, ARNS, RNSS (Earth-to-space) band

3.1.4 Test Procedures

1. Configure the GNSS signal generator to simulate those GNSS signals from Table 4-1 declared as supported by the GUE, with power levels and other details as specified in Table B-1.
2. With the Blocking signal switched off, the EUT shall be given sufficient time to acquire all simulated satellites from the declared GNSS system(s).
3. Record the baseline C/N0 value reported by the EUT. Sufficient filtering shall be used to obtain a stable value. C/N0 may be averaged across all the satellites in view for each constellation. However, C/N0 shall not be averaged across satellite signals in different constellations. For a multi-GNSS EUT, there shall be a separate C/N0 value recorded for each constellation supported.
4. The blocking signal generator shall be configured to generate the signal defined in Table 4-4, at the first test point centre frequency and signal power level as specified in Table 4-2.
5. The blocking signal shall be switched on, and the EUT's effective C/N0 recorded as in step 3 to measure degradation with respect to the baseline recorded in step 3.
6. Test point Pass/Fail Criteria: If the C/N0 degradation from step 5 does not exceed the value in equation (4-1), then this test point is set to "pass". If the C/N0 degradation exceeds the value in equation (4-1), then this test point is set to "fail." For a multi-GNSS EUT, there will be a separate pass/fail determination for each GNSS supported. If the C/N0 degradation exceeds the value in equation (4-1) for any supported GNSS, then this test point is set to "fail".
7. Steps 1 through 6 shall be repeated for all test point centre frequencies (and associated signal power level) specified in Table 4-2.
8. If the EUT passes the C/N0 degradation test for all test points for all GNSS constellations, the EUT shall be deemed to "pass". If the C/N0 degradation test fails for any constellation at any of the test points, the EUT shall be deemed to "fail".

3.1.5 Test Setup

Conducted measurement setup



3.1.6 Test Result of Receiver Blocking

Conducted or radiated testing:

■ Conducted

□ Radiated

Final test results for 1559 MHz to 1610 MHz RNSS band:

■ Pass □ Fail

Final test results for 1164 MHz to 1300 MHz RNSS band:

□ Pass □ Fail ■ N/A

Please refer to Appendix A for the detailed test data.

3.2 Receiver Spurious Emissions

3.2.1 Requirement

The spurious emissions of the GUE shall not exceed the values given in Table 4-5.

In case of a GUE with an external antenna connector, these limits apply to emissions at the antenna port (conducted). For emissions radiated by the cabinet or for emissions radiated by a GUE with an integral antenna (without an antenna connector), these limits are e.r.p. for emissions up to 1 GHz and e.i.r.p. for emissions above 1 GHz.

EN 303 413 Table 4-5: Spurious emission limits

Frequency range	Maximum power	Bandwidth
30 MHz to 1 GHz	-57 dBm	100 kHz
1 GHz to 8.3 GHz	-47 dBm	1 MHz

3.2.2 Measuring Instruments

The measurement equipment used complied with the requirement of standard as listed the section 4 Listing of Measuring Equipment of this test report.

3.2.3 Test Procedures

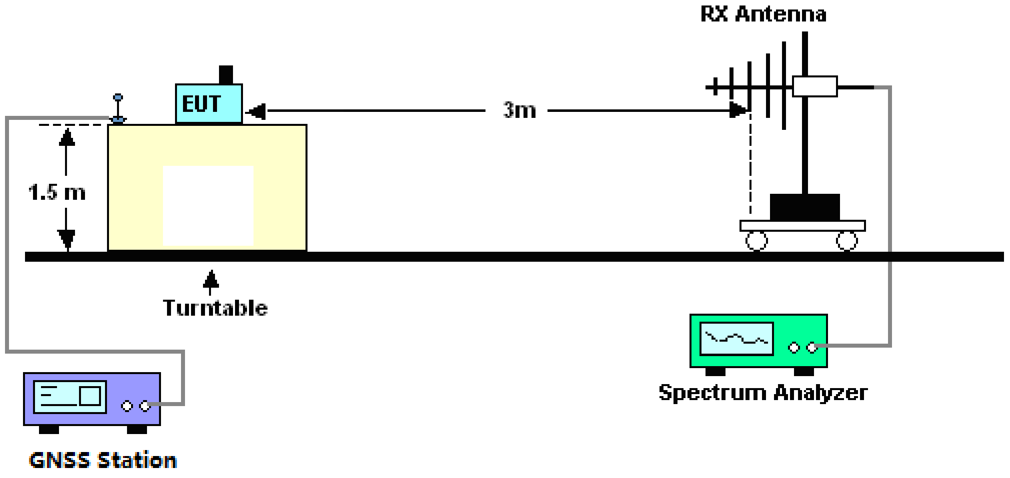
1. The testing was performed in shielded Fully Anechoic Chamber (FAR).
2. The EUT was placed on a turntable with 1.5m height.
3. The receiving antenna with horizontal and vertical polarization is 3m away from EUT and keeps the antenna height at 1.5m.
4. Setting EUT in continuous Rx.
5. The table was rotated to search the highest radiation.
6. Test Results Explanation:

   Level (dBm) = Read Level (dBm) + Factor (dB)

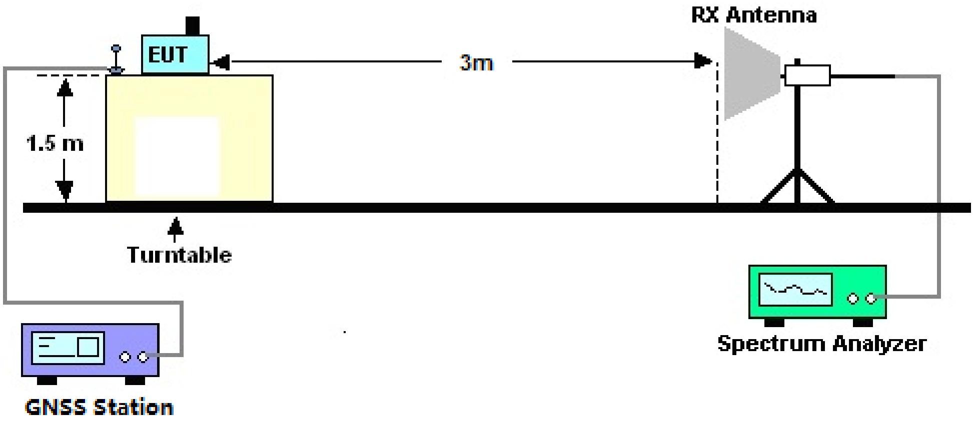
   Over Limit (dB) = Level (dBm) – Limit Line (dBm)

3.2.4 Test Setup

<Below 1GHz>



<Above 1GHz>



3.2.5 Test Result of receiver spurious emissions

GNSS signal(s) present or absent:

■ Present □ Absent

Receiver Spurious Emissions test result:

■ Pass □ Fail □ N/A

Please refer to Appendix A for the detailed test data.

4. List of Measuring Equipment

Instrument	Manufacturer	Model No.	Serial No.	Characteristics	Calibration Date	Test Date	Due Date	Remark
Signal and spectrum analyzer	Rohde & Schwarz	FSV3044	101177	10Hz-44GHz	Jul. 03, 2025	Mar. 14, 2026	Jul. 02, 2026	Radiation (05CH02-KS)
Log-periodic antenna	R&S	HL562	100446	30MHz-3GHz	Jan. 09, 2026	Mar. 14, 2026	Jan. 08, 2027	Radiation (05CH02-KS)
Horn Antenna	R&S	HF906	100485	1GHz~18GHz	Feb. 07, 2026	Mar. 14, 2026	Feb. 06, 2027	Radiation (05CH02-KS)
SHF-EHF Horn	Com-Power	AH-840	101116	18GHz~40GHz	Oct. 18, 2025	Mar. 14, 2026	Oct. 17, 2026	Radiation (05CH02-KS)
Amplifier	Keysight	83017A	MY53270360	500MHz~26.5G Hz	Apr. 09, 2025	Mar. 14, 2026	Apr. 08, 2026	Radiation (05CH02-KS)
Amplifier	SONOMA	310N	372171	9KHz ~1GHZ	Dec. 24, 2025	Mar. 14, 2026	Dec. 23, 2026	Radiation (05CH02-KS)
Amplifier	EM	EM18G40GA	060851	18~40GHz	Dec. 26, 2025	Mar. 14, 2026	Dec. 25, 2026	Radiation (05CH02-KS)
Amplifier	EM	EM01G18GA	060839	1GHz-18GHz	Sep. 19, 2025	Mar. 14, 2026	Sep. 18, 2026	Radiation (05CH02-KS)
Turn Table	MF	MF7802	N/A	0~360 degree	NCR	Mar. 14, 2026	NCR	Radiation (05CH02-KS)
Antenna Mast	MF	MF7802	N/A	1 m~4 m	NCR	Mar. 14, 2026	NCR	Radiation (05CH02-KS)
Vector Signal Generator(BS)	R&S	SMBV100A	260891	9kHz~3.2GHz	Jul. 02, 2025	May 19, 2026	Jul. 01, 2026	Conducted (DFS01-KS)
GNSS STATION	Spectracom	GSG-6 Series	201883	GNSS STATION	Oct. 11, 2025	May 19, 2026	Oct. 10, 2026	Conducted (DFS01-KS)
Vector Signal Generator (Interferer)	R&S	SMBV100A	258305	9kHz~6GHz	Jan. 02, 2026	May 19, 2026	Jan. 01, 2027	Conducted (DFS01-KS)
Combiner	MTJ Cooperation	MTJ7114-M	N/A	0.5GHz~18GHz	NCR	May 19, 2026	NCR	Conducted (DFS01-KS)

NCR: No Calibration Required

5. Measurement Uncertainty

Test Item	Frequency Range	Uncertainty
Radiated spurious emissions	25MHz ~ 1GHz	±2.54dB
	1GHz ~ 18GHz	±2.54dB

----------- THE END -----------

Appendix A. Test Results

A.1 Receiver Blocking test results

Test results for 1559 MHz to 1610 MHz RNSS band:

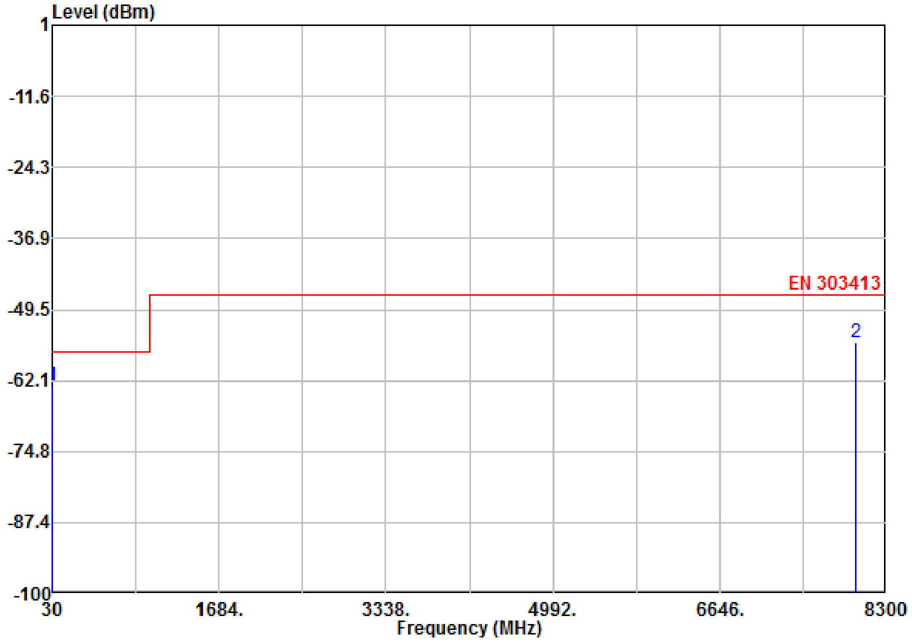
Test Mode :	GPS L1C/A Rx Mode GPS L1C Rx Mode BDS B1I Rx Mode GLONASS Rx Mode	**Temperature :**	20-25℃
Test Engineer :	Han Lei	**Relative Humidity :**	40~55%

Frequency band (MHz)	**Test point centre frequency (MHz)**	**Adjacent frequency signal power level (dBm)**	**Measured C/No (dB)**			
			No interfering signal	With interfering signal	Decrease of C/No	Decrease <= 1 dB
1518- 1525	1524	-65	38	38	0	BDS B1I ■ Pass □ Fail □ N/A
						BDS B1C □ Pass □ Fail □ N/A
			40	40	0	Galileo E1 ■ Pass □ Fail □ N/A
						GLONASS □ Pass □ Fail □ N/A
			44	44	0	GPS L1C ■ Pass □ Fail □ N/A
			43	43	0	GPS L1C/A ■ Pass □ Fail □ N/A
						SBAS L1 □ Pass □ Fail □ N/A
1525-1549	1548	-95	38	38	0	BDS B1I ■ Pass □ Fail □ N/A
						BDS B1C □ Pass □ Fail □ N/A
			40	40	0	Galileo E1 ■ Pass □ Fail □ N/A
						GLONASS □ Pass □ Fail □ N/A
			44	44	0	GPS L1C ■ Pass □ Fail □ N/A
			43	43	0	GPS L1C/A ■ Pass □ Fail □ N/A
						SBAS L1 □ Pass □ Fail □ N/A

1549-1559	1554	-105	38	38	0	BDS B1I ■ Pass □ Fail □ N/A
						BDS B1C □ Pass □ Fail □ N/A
			40	40	0	Galileo E1 ■ Pass □ Fail □ N/A
						GLONASS □ Pass □ Fail □ N/A
			44	44	0	GPS L1C ■ Pass □ Fail □ N/A
			43	43	0	GPS L1C/A ■ Pass □ Fail □ N/A
						SBAS L1 □ Pass □ Fail □ N/A
1610-1626	1615	-105	38	38	0	BDS B1I ■ Pass □ Fail □ N/A
						BDS B1C □ Pass □ Fail □ N/A
			40	40	0	Galileo E1 ■ Pass □ Fail □ N/A
						GLONASS □ Pass □ Fail □ N/A
			44	44	0	GPS L1C ■ Pass □ Fail □ N/A
			43	43	0	GPS L1C/A ■ Pass □ Fail □ N/A
						SBAS L1 □ Pass □ Fail □ N/A
1626- 1640	1627	-85	38	38	0	BDS B1I ■ Pass □ Fail □ N/A
						BDS B1C □ Pass □ Fail □ N/A
			40	40	0	Galileo E1 ■ Pass □ Fail □ N/A
						GLONASS □ Pass □ Fail □ N/A
			44	44	0	GPS L1C ■ Pass □ Fail □ N/A
			43	43	0	GPS L1C/A ■ Pass □ Fail □ N/A
						SBAS L1 □ Pass □ Fail □ N/A

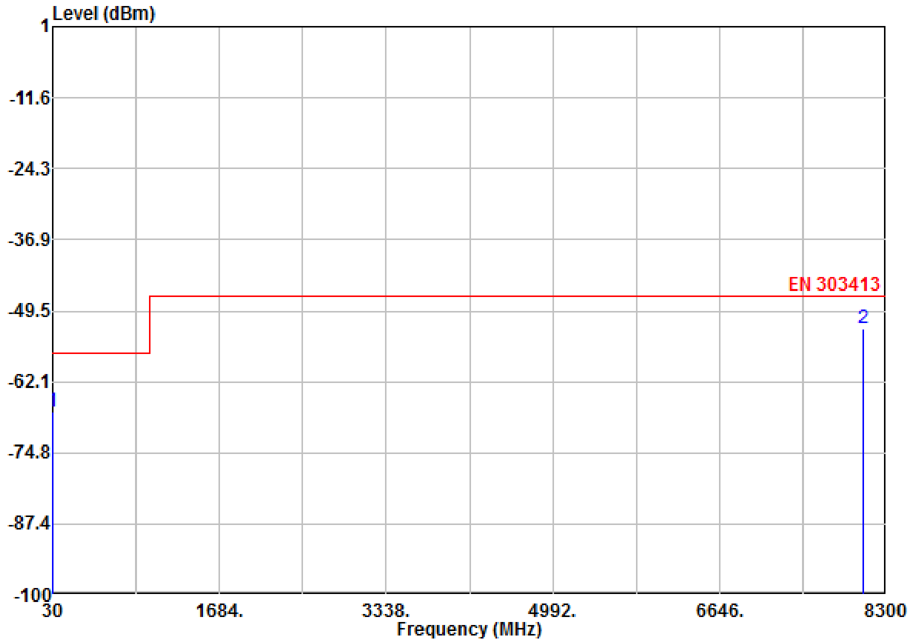
A.2 Receiver spurious emissions

Test Mode :	GNSS RX	Temperature :	22~23℃
Test Engineer :	Juggler Zhang	Relative Humidity :	42~43%
		Polarization :	Horizontal



	Result	Freq	Level	Over Limit	Limit Line	Read Level	Factor	Pol/Phase
		MHz	dBm	dB	dBm	dBm	dB	
1	@	31.94	-62.99	-5.99	-57.00	-66.08	3.09	Horizontal
2		7995.82	-55.28	-8.28	-47.00	-48.61	-6.67	Horizontal

Test Mode :	GNSS RX	Temperature :	22~23℃
Test Engineer :	Juggler Zhang	Relative Humidity :	42~43%
		Polarization :	Vertical



Result	Freq	Level	Over Limit	Limit Line	Read Level	Factor	Pol/Phase
	MHz	dBm	dB	dBm	dBm	dB	
1	31.94	-67.44	-10.44	-57.00	-68.50	1.06	Vertical
2 @	8064.36	-52.64	-5.64	-47.00	-46.74	-5.90	Vertical

Appendix B. Setup Photographs

Front View

(Radiated)

Front View

(Conducted)

———THE END———